Design and Verification of LTSSM in USB 3.0 Link Layer using System Verilog

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***Abstract*— The proposed design includes the Link Training and Status State Machine (LTSSM). The LTSSM has both the upstream port and downstream port. The transition between all the 12 link states has been designed and verified. Complex scenarios are covered with multiple cover groups with many cover points. The RTL behavior of the LTSSM in Link Layer is verified using System Verilog and its verification is carried out in Mentor Graphics tool. The proposed verification environment includes the constrained randomization, functional coverage and code coverage. In the proposed system, DUT is subjected to the verification with various coverage metrics such as code coverage, functional coverage, etc. The functional coverage obtained 87.5 percentage by using cover groups.**

Keywords—USB 3.0; LTSSM; Verification Environment; System Verilog; Coverage.

I. INTRODUCTION

The Universal Serial Bus (USB) is developed in the middle of 1990s that gives the definition for the connectors cables and communications formulae used in a bus for power supply, connection and communication between electronic devices and computers [1].

It is used to provide communication between a host controller (usually a personal computer) and devices. Ajay Bhatt invented this while working for Intel. It has effectively replaced a variety of interfaces like parallel ports and serial. USB can connect computer peripherals such as external hard drives, flash drives, Network Adapters mouse, keyboards, personal media players, printers and digital cameras [3]. Standard connection method used for many of those devices is USB.

Computers used parallel and serial ports to plug devices for transferring data before USB were invented. Expansion cards and custom drivers were often required to connect the devices. Data are transferred using parallel port at approximately 110 kbps, whereas serial ports ranged from 110 to more than 400 kbps.

Versions of USB Specification:

- USB 0.7: Released in November 1994.
- USB 0.8: Released in December 1994.
- USB 0.9: Released in April 1995.
- USB 1.0: Released in January 1996.
- USB 1.1: Released in September 1998.
- USB 2.0: Released in April 2000.
- USB 3.0: Released in November 2008.

The paper outline is explained as follows. Section II explains about USB 3.0 Link layer and Section III explains about the LTSSM and the pin diagram of LTSSM. In Section IV, USB 3.0 LTSSM Link States which includes SS.Disabled, SS.Inactive, Rx.Detect, Polling, Compliance, U0, U1, U2, U3, Recovery, Loopback and Hot Reset working are described. The Verification Environment for LTSSM is detailed in Section IV. Next in Section V, the simulation results are given. Finally conclusion is drawn.

II. USB 3.0 LINK LAYER

The Link Layer is used to manage traffic between the two links which are connected to each other. It manages the flow of data from port to port between the device and the host. Packets are used to carry different information and data in the link layer between the device and host [2].

Link commands are used to define the communication between the two link partners. The link layer also facilitates the link testing/debugging, link training and link power management. The USB 3.0 Layout diagram is shown in Fig.1.

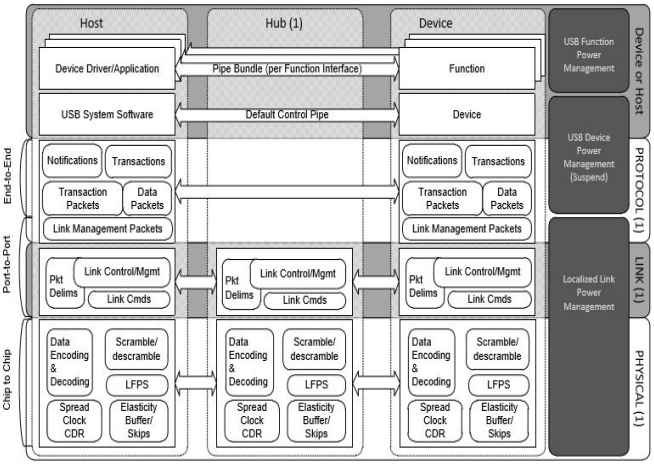


Fig. 1. USB Layout Diagram

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The Link Layer functions consist of:

- Effective power management for 4 link power state (U0, U1, U2, and U3).
- Packet and link command formation.
- Link training symbol lock and Rx-equalization.
- Packet header formation.
- Different types of error handling.

III. USB 3.0 LTSSM

Link Training and Status State Machine (LTSSM) is a state machine defined for power management and connectivity. LTSSM consists of 12 different link states with various functionalities. The four operational link states are U0, U1, U2, and U3. The link states Polling, Rx.Detect, Hot Reset and Recovery are introduced for link training and initialization. The other two states Compliance Mode and Loopback are introduced for transmitter compliance test and bit error test [5]. The State diagram for LTSSM is shown in Fig. 2.

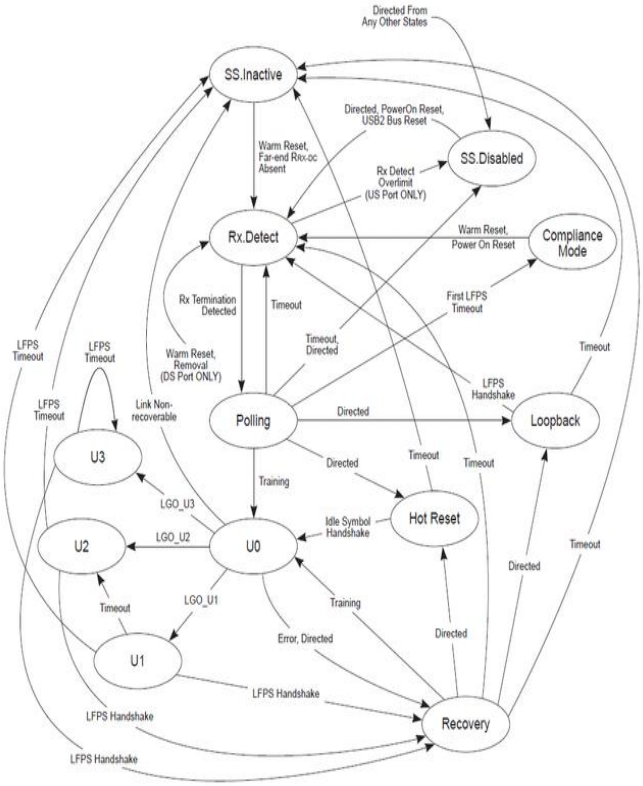


Fig. 2. LTSSM State Diagram

SS.Inactive is a link error state which is found in a non-operable state. In this state software intervention will be needed. SuperSpeed connectivity is disabled and in this state, the link is operated using USB 2.0 mode. In LTSSM state machine, directed represents upper layer mechanisms. The Pin diagram for LTSSM is shown in Fig. 3.

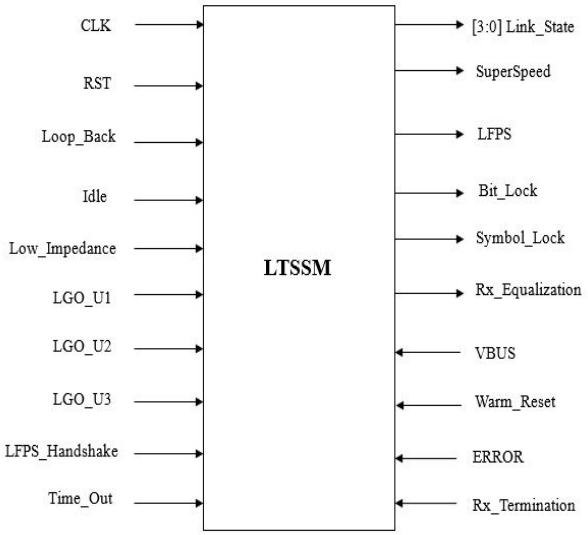


Fig. 3. LTSSM Pin Diagram

IV. USB 3.0 LTSSM STATES

The USB 3.0 LTSSM architecture consists of the following 12 states and also includes the substates.

A. SS.Disabled

SS.Disabled is a state where SuperSpeed connectivity is disabled. SS.Disabled does not contain any sub-state.

B. SS.Inactive

SS.Inactive is a state where the USB is non-operable. SS.Inactive has two types of substate machines as shown in the Fig. 4.

- SS.Inactive.Disconnect.Detect
- SS.Inactive.Quiet

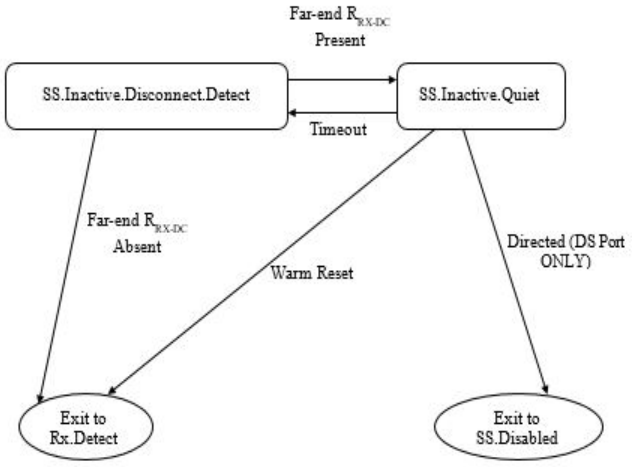


Fig. 4. SS.Inactive Sub-State Machine

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C. Rx.Detect

Rx.Detect is the power on state of the LTSSM. It is also the state for a downstream port on issue of a Warm Reset and the state for an upstream port on detection of a Warm Reset. Rx.Detect is used to detect the far-end receiver termination.

Rx.Detect has three substate machine as shown in the below Fig. 5.

- Rx.Detect.Reset
- Rx.Detect.Active
- Rx.Detect.Quiet

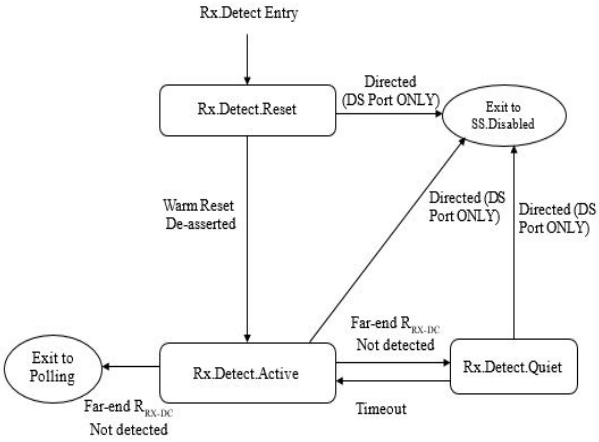


Fig. 5. Rx.Detect Sub-State Machine

D. Polling

Polling is a state for link training. Before link training is started, Polling.LFPS handshake will take place between two ports. Symbol lock, Rx equalization and Bit lock are achieved using TS1, TS2 and TSEQ ordered sets.

Polling has five types of substate machine as shown in the Fig. 6.

- Polling.LFPS
- Polling.RxEQ
- Polling.Active
- Polling.Configuration
- Polling.Idle

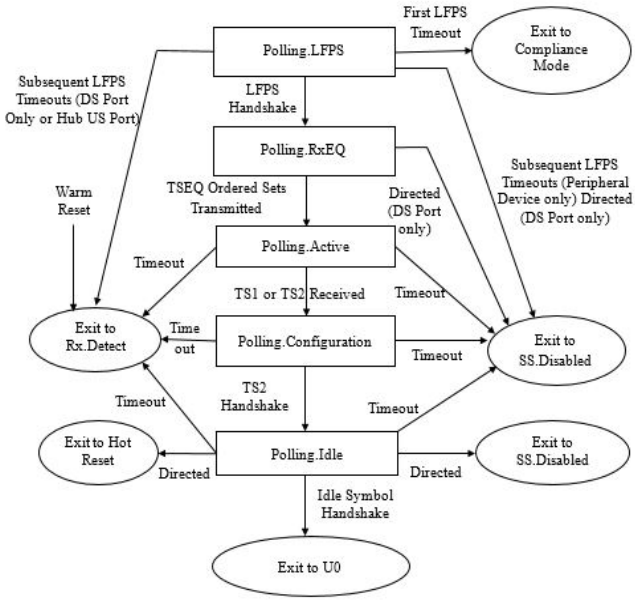


Fig. 6. Polling Sub-State Machine

E. Compliance Mode

Compliance Mode is used to test the voltage and timing specifications. Compliance Mode does not include any kind of substate machines.

F. U0

U0 is the normal operational state where packets can be transmitted and received. U0 is a state where the SuperSpeed link is enabled. U0 does not contain any substate machines.

G. U1

U1 is a low power link state. In this state packet transfer is not carried out. The SuperSpeed link connectivity can be disabled for saving the link power. U1 does not have any kind of substates. Transitions from U1 state to other states are shown in Fig.7.

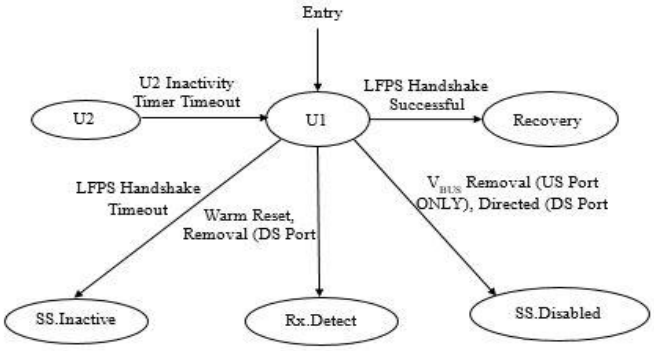


Fig. 7. Transitions to other states from U1

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H. U2

U2 is a link state where more power saving opportunities are allowed with an increased exit latency. U2 does not contain any kind of substates. The transitions from U2 to other states are shown in Fig.8.

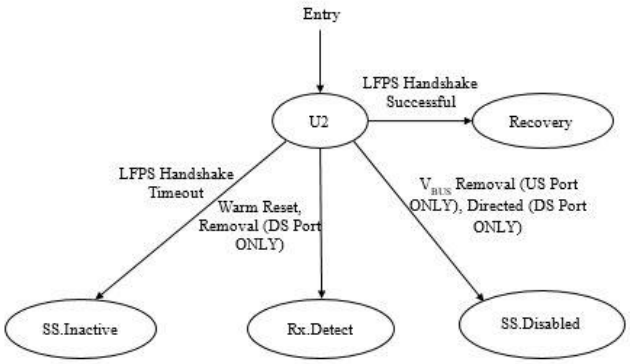


Fig. 8. Transitions to other states from U2

I. U3

U3 is a link suspend state where aggressive power saving opportunities are possible. Significant link as well as device powers are saved. U3 does not contain any kind of substates. Transitions from U3 to other states are shown in Fig.9.

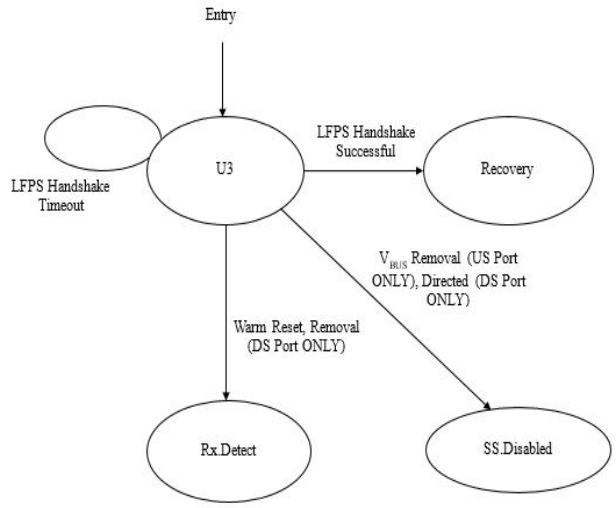


Fig. 9. Transitions to other states from U3

J. Recovery

The Recovery is a link state used for retraining of the link or to perform Hot Reset or to switch to Loopback mode. For retraining purpose, the link partners will not train the receiver equalizers, it will maintain the last receiver configuration.

Recovery has three types of substate machine as shown in the below Fig. 10.

- Recovery.Active
- Recovery.Configuration
- Recovery.Idle

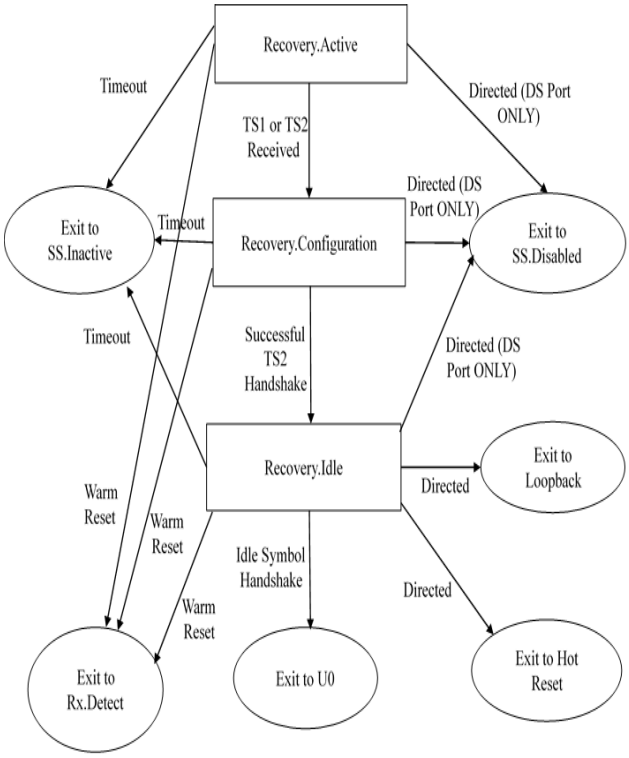


Fig. 10. Recovery Sub-State Machine

K. Loopback

Loopback is used for test and fault isolation. Loopback includes BERT (Bit Error Rate Test). A loopback master is the port which will request loopback. A loopback slave will retransmit the symbols that are received from loopback master [1].

Loopback has two types of substate machine as shown in the below Fig. 11.

- Loopback.Active
- Loopback.Exit

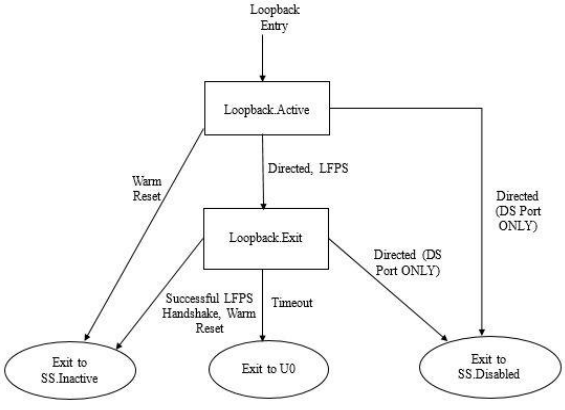


Fig. 11. Loopback Sub-State Machine

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L. Hot Reset

Only downstream port is used to initiate a Hot Reset. When downstream port initiates reset, it will transmit TS2 ordered sets. The upstream port will respond by sending the TS2 ordered sets.

Hot Reset has two types of substate machine as shown in the below Fig. 12.

- Hot Reset.Active
- Hot Reset.Exit

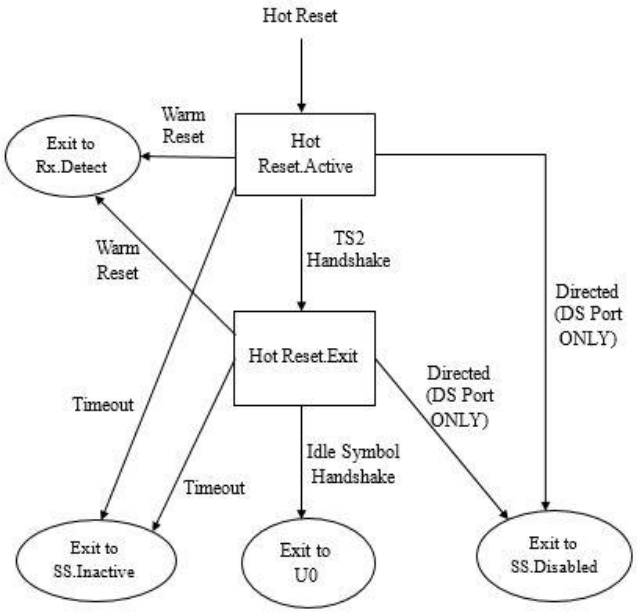


Fig. 12. Hot Reset Sub-State Machine

V. LTSSM Verification Environment

The well-structured verification environment is required for more complex design as shown in Fig. 13. The fundamental method which is used for verification is Simulation [4].

The Verification Plan will define which test case has to be tested. For success of verification project, verification plan and its accurate implementation is must.

System Verilog is more helpful for the verification process. Object Oriented Programming (OOP) technique is applied in verification environment of System Verilog.

The main purpose of verification engineer is to check whether the device is performing the task exactly as it is in the specification. System Verilog is used for the verification purpose. The two different verification levels are IP level verification and Top level verification. Based on the design specification, the verification environment is to be created. Constrained random stimulus, Code Coverage and Functional Coverage results can be obtained [10].

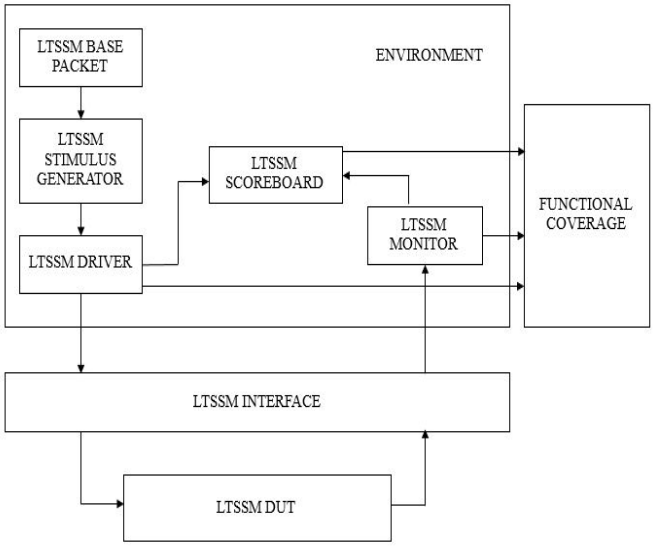


Fig. 13. System Verilog Architecture

VI. Simulation Results And Discussion

In the proposed system, simulated waveforms and coverage reports for the functionality verification of USB 3.0 LTSSM are described below. The standard of using the system Verilog based verification environment shows best result compared to other verification environments.

The LTSSM states SS.Disabled, SS.Inactive, Rx.Detect, Polling, Compliance Mode, U0, U1, U2, U3, Recovery, Loopback and Hot Reset transitions are occurred based on the inputs such as LGOU1, LGOU2, LGOU3, LFPS Handshake, Warm Reset, Error, Low Impedance Rx_Termination, Idle, Timeout, Loop Back and VBUS is shown below in the Fig. 14.

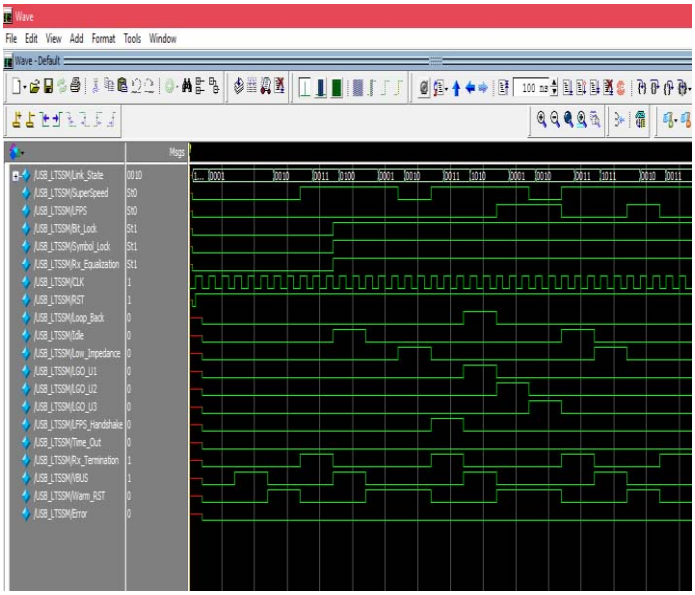


Fig. 14. Simulation result for LTSSM

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A. Verification Environment Results

The Constrained Randomization Technique is applied for the inputs of the LTSSM and the verification environment is developed by using System Verilog.

The comparison result is obtained from the scoreboard. In the Scoreboard, the link states in the DUT will be compared with the reference model part created in the scoreboard and it is verified that the actual link state and the expected link state in the reference model obtained are same. The comparison result for the verification environment by using System Verilog is shown in Fig. 15.

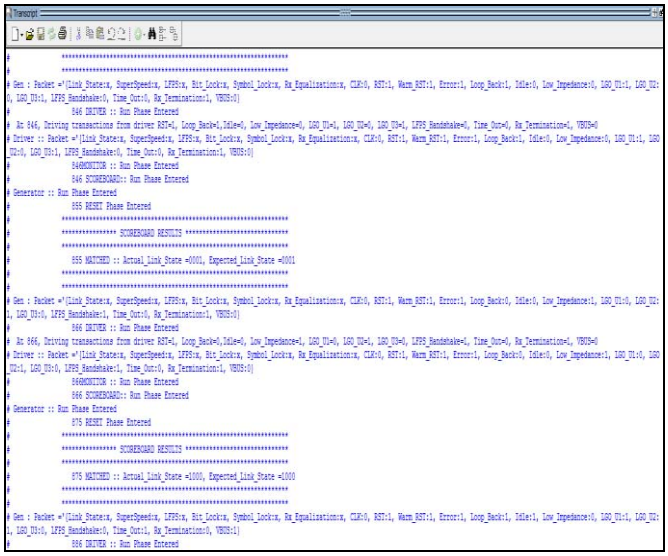


Fig. 15. Scoreboard Results

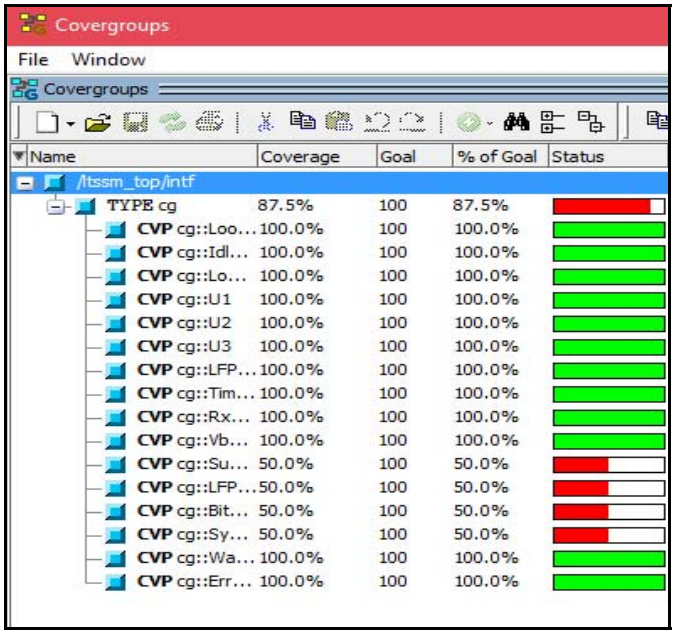


Fig. 16. Functional Coverage Result for LTSSM

The Functional Coverage for the LTSSM is obtained as 87.5% by using the cover groups. In the test plan, if all the scenarios are tested then 100% Functional Coverage will be obtained. The Functional Coverage for LFPS, Bit Lock, Symbol Lock and SuperSpeed obtained is 50% and so the total functional coverage obtained is 87.5% and it is shown in Fig. 16.

VII. CONCLUSION

The USB 3.0 LTSSM was successfully designed using Verilog and Simulated. The Verification Environment was created using System Verilog. The proposed verification environment comprised of interface, generator, monitor, driver and scoreboard were implemented by using OOP concepts. The functionality of the design is verified by the System Verilog Verification Environment. The constrained randomization technique is applied for the design and verification environment. In scoreboard the actual state of the LTSSM matches with the expected state of LTSSM. The functional coverage obtained was 87.5% by using the cover groups. This link state in LTSSM can be used as an input for link layer.

REFERENCES

[1] Universal Serial Bus 3.0 Specification Revision 1, November12, 2008.

[2] Universal Serial Bus Specification Revision 2.0, April 2000.

[3] USB 3.1 Specification Revision 1.0, 26 July 2013.

[4] Jammu Soma Sekhar, Maley Srinivasa Raor and K.Venu Gopal, "VLSI Implementation of Transmitting Part of Link Layer in Super Speed USB," Journal of International Academic Research for Multidisciplinary, vol.2, Issue 9, pp.400-407, October 2014.

[5] Durgesh Kumar Sahu, Shweta Singh, and Nitin Meena, "Implementation of Link Layer of USB 3.1", International Journal of Innovative Research in Computer and Communication Engineering, vol.2, Issue 1, pp. 2593-2598, January 2014.

[6] Monika S, and Shruthi G, "Study and Implementation of Physical Layer Coding Used in Super Speed USB", International Journal of Scientific Engineering and Research (IJSER), vol.2, Issue 1, pp.52-55, January 2014.

[7] Krunal Kapadiya, "Verification of USB 3.0 Device IP using Universal Verification Methodology," International Journal of Engineering Research and Applications (IJJERA), vol.3, Issue.3, pp. 551-556, June 2013.

[8] K.V.Suresh Babu, A.S.Srinivasarao, and D.Srinivasa Rao, "VLSI Implementation of Physical Layer Coding used in Super Speed USB using Verilog," Global Journal of Advanced Engineering Technologies, vol.1, Issue 4, pp.218-222, 2012.

[9] Arpit Patel, "UVM Based Verification Environment for USB 3.0 Physical layer and LTSSM of Link layer," International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC), vol.3, Issue 5, pp.2811-2815, May 2015.

[10] K.Gavaskar, and M.Sukhanya, "Design and Verification of I2C Master Controller using System Verilog," International Journal of Trend in Research and Development (IJTRD), ISSN: 2394-9333, December 2016.